

Practical – 1 Create a real-world **Online Registration Form (Event App)** using Flutter

Create a registration form with:

- Full Name, Email, Phone (TextFormField)
- Gender (RadioListTile)
- City (DropDownButtonFormField)
- Accept Terms & Conditions (Checkbox)
- Submit Button

Step 1: Create Flutter Project

Step 2: Write a code in lib/main.dart with the following code:

```
import 'package:flutter/material.dart';
```

```
void main() {  
  runApp(MaterialApp(  
    debugShowCheckedModeBanner: false,  
    home: RegistrationForm(),  
  ));  
}
```

```
class RegistrationForm extends StatefulWidget {  
  @override  
  _RegistrationFormState createState() => _RegistrationFormState();  
}
```

```
class _RegistrationFormState extends State<RegistrationForm> {  
  final _formKey = GlobalKey<FormState>();  
}
```

String? _name;

String? _email;

String? _phone;

String? _gender;

String? _selectedCity;

bool _agree = false;

List<String> _cities = ['Ahmedabad', 'Surat', 'Rajkot', 'Vadodara',
'Gandhinagar'];

@override

Widget build(BuildContext context) {

 return Scaffold(

 appBar: AppBar(

 title: Text('Event Registration Form'),

),

 body: SafeArea(

 child: Padding(

 padding: EdgeInsets.all(16.0),

 child: Form(

 key: _formKey,

 child: ListView(

 children: [

```
/// Full Name
```

```
TextFormField(  
  decoration: InputDecoration(labelText: 'Full Name'),  
  validator: (value) {  
    if (value == null || value.isEmpty) {  
      return 'Please enter your full name';  
    }  
    return null;  
  },  
  onSave: (value) => _name = value,  
)
```

```
/// Email
```

```
TextFormField(  
  decoration: InputDecoration(labelText: 'Email'),  
  keyboardType: TextInputType.emailAddress,  
  validator: (value) {  
    if (value == null || value.isEmpty || !value.contains('@')) {  
      return 'Enter valid email';  
    }  
    return null;  
  },  
  onSave: (value) => _email = value,
```

),

/// Phone

```
TextFormField(  
  decoration: InputDecoration(labelText: 'Phone Number'),  
  keyboardType: TextInputType.phone,  
  validator: (value) {  
    if (value == null || value.length != 10) {  
      return 'Enter 10-digit phone number';  
    }  
    return null;  
  },  
  onSaved: (value) => _phone = value,  
),
```

SizedBox(height: 20),

/// Gender (Radio)

```
Text('Gender:', style: TextStyle(fontWeight: FontWeight.bold)),  
RadioListTile<String>(  
  title: Text('Male'),  
  value: 'Male',  
  groupValue: _gender,  
  onChanged: (value) {
```

```
        setState(() => _gender = value);
    },
),
```

```
RadioListTile<String>{
    title: Text('Female'),
    value: 'Female',
    groupValue: _gender,
    onChanged: (value) {
        setState(() => _gender = value);
    },
),
```

```
/// City (Dropdown)
```

```
DropdownButtonFormField<String>{
    decoration: InputDecoration(labelText: 'Select City'),
    items: _cities
        .map((city) => DropdownMenuItem(value: city, child: Text(city)))
        .toList(),
    onChanged: (value) {
        setState(() => _selectedCity = value);
    },
    validator: (value) => value == null ? 'Please select a city' : null,
),
```

```
/// Accept Terms (Checkbox)

CheckboxListTile(

  title: Text('I agree to the Terms & Conditions'),

  value: _agree,

  onChanged: (value) {

    setState(() => _agree = value ?? false);

  },

  controlAffinity: ListTileControlAffinity.leading,

),

SizedBox(height: 20),

/// Submit Button

ElevatedButton(

  child: Text('Submit'),

  onPressed: () {

    if (_formKey.currentState!.validate()) {

      if (!_agree) {

        ScaffoldMessenger.of(context).showSnackBar(

          SnackBar(content: Text('Please accept the terms')),

        );

        return;

      }

    }

  }

)
```

```
_formKey.currentState!.save();
```

```
showDialog(  
  context: context,  
  builder: (_) => AlertDialog(  
    title: Text('Registration Successful'),  
    content: Text('Thank you, $_name!\n'  
      'Email: $_email\n'  
      'Phone: $_phone\n'  
      'Gender: $_gender\n'  
      'City: $_selectedCity'),  
    actions: [  
      TextButton(  
        onPressed: () => Navigator.pop(context),  
        child: Text('OK'),  
      )  
    ],  
  ),  
);  
}  
},  
)  
],  
)
```

```

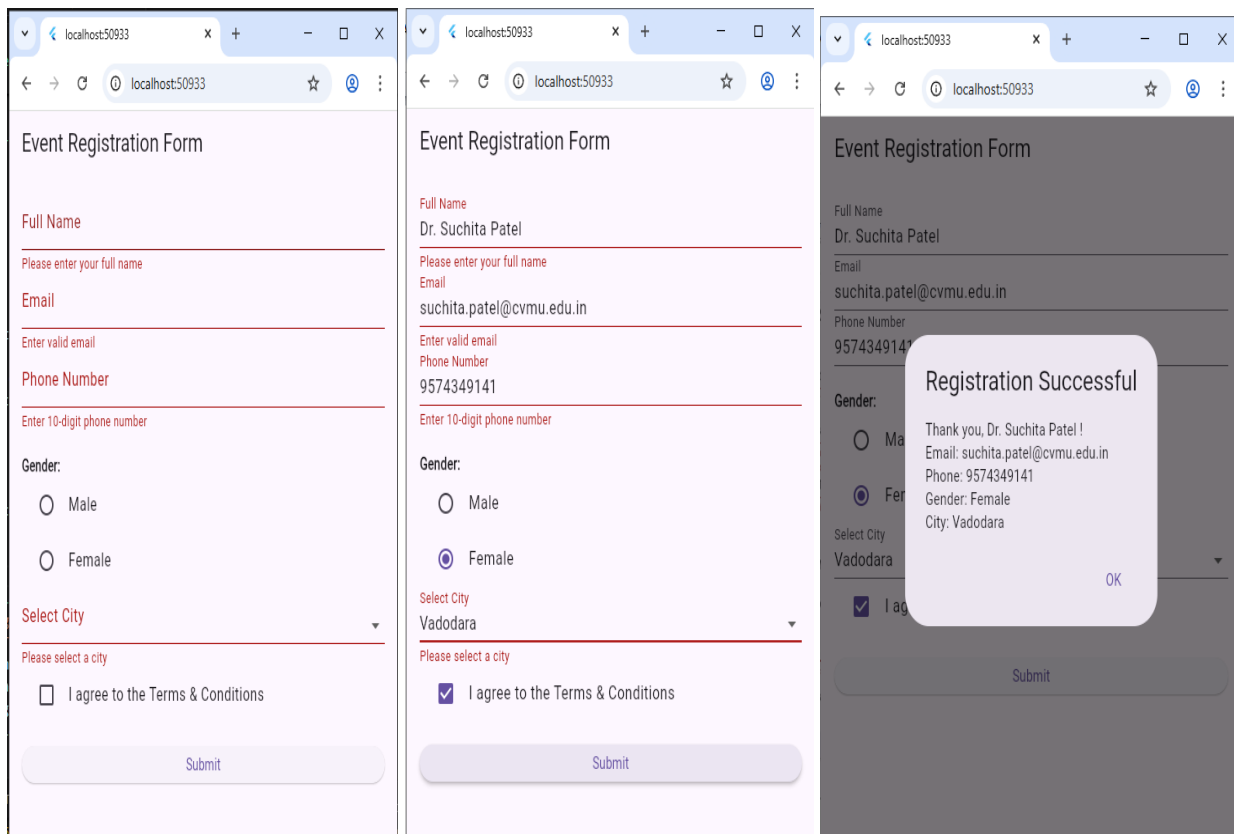
    ),
    ),
    ),
    );
}
}

```

Step – 3 Run Your App.

Flutter run

Output:



Practical – 2 : Build a **Car Booking App (Form-based)** in **Flutter**

Widgets used:

- TextFormField: Name, Pickup & Drop Location
- DropdownButtonFormField: Car Type
- RadioListTile: Payment Option
- CheckboxListTile: Add insurance, GPS
- Image.asset: Show car images

Real-world use: Ride-booking and rental car reservation apps.

Step 1: Create Flutter Project

```
flutter create car_booking_app
```

```
cd car_booking_app
```

Step 2: Add Car Images

Put car images in an assets/ folder in your root directory like:

```
assets/car1.png
```

```
assets/car2.png
```

Update your pubspec.yaml:

```
flutter:
```

```
  assets:
```

```
    - assets/car1.png
```

```
    - assets/car2.png
```

Step 3: Write a code in lib/main.dart with the Full Code Below

```
import 'package:flutter/material.dart';
```

```
void main() {
```

```
runApp(MaterialApp(  
  debugShowCheckedModeBanner: false,  
  home: CarBookingForm(),  
));  
}
```

```
class CarBookingForm extends StatefulWidget {  
  @override  
  _CarBookingFormState createState() => _CarBookingFormState();  
}
```

```
class _CarBookingFormState extends State<CarBookingForm> {  
  final _formKey = GlobalKey<FormState>();  
  
  String? _name;  
  String? _pickupLocation;  
  String? _dropLocation;  
  String? _carType;  
  String? _paymentOption;  
  
  bool _addInsurance = false;  
  bool _addGPS = false;  
  
  final List<String> _carTypes = ['Sedan', 'SUV', 'Hatchback', 'Luxury'];
```

@override

Widget build(BuildContext context) {

 return Scaffold(

 appBar: AppBar(title: Text('Car Booking Form')),

 body: SafeArea(

 child: Padding(

 padding: EdgeInsets.all(16),

 child: Form(

 key: _formKey,

 child: ListView(

 children: [

 /// Show Car Image

 Center(

 child: Image.asset(

 'assets/car1.png',

 height: 150,

),

),

 SizedBox(height: 20),

 /// Name

 TextFormField(

```
        decoration: InputDecoration(labelText: 'Full Name'),
        validator: (value) => value == null || value.isEmpty ? 'Enter name' :
null,
        onSave: (value) => _name = value,
    ),
```

```
    /// Pickup Location
    TextFormField(
        decoration: InputDecoration(labelText: 'Pickup Location'),
        validator: (value) => value == null || value.isEmpty ? 'Enter pickup
location' : null,
        onSave: (value) => _pickupLocation = value,
    ),
```

```
    /// Drop Location
    TextFormField(
        decoration: InputDecoration(labelText: 'Drop Location'),
        validator: (value) => value == null || value.isEmpty ? 'Enter drop
location' : null,
        onSave: (value) => _dropLocation = value,
    ),
```

```
    SizedBox(height: 20),
```

```
    /// Car Type Dropdown
```

```
DropdownButtonFormField<String>(  
  decoration: InputDecoration(labelText: 'Select Car Type'),  
  items: _carTypes.map((car) {  
    return DropdownMenuItem(value: car, child: Text(car));  
  }).toList(),  
  onChanged: (value) => setState(() => _carType = value),  
  validator: (value) => value == null ? 'Please select a car type' : null,  
)
```

```
SizedBox(height: 20),
```

```
/// Payment Option Radio
```

```
Text('Payment Option:', style: TextStyle(fontWeight: FontWeight.bold)),
```

```
RadioListTile<String>(  
  title: Text('Cash'),  
  value: 'Cash',  
  groupValue: _paymentOption,  
  onChanged: (value) => setState(() => _paymentOption = value),  
)
```

```
RadioListTile<String>(  
  title: Text('Online'),  
  value: 'Online',  
  groupValue: _paymentOption,  
  onChanged: (value) => setState(() => _paymentOption = value),
```

```
),
```

```
/// Add-ons: Insurance
```

```
CheckboxListTile(
```

```
  title: Text('Add Insurance'),
```

```
  value: _addInsurance,
```

```
  onChanged: (value) => setState(() => _addInsurance = value ?? false),
```

```
  controlAffinity: ListTileControlAffinity.leading,
```

```
),
```

```
/// Add-ons: GPS
```

```
CheckboxListTile(
```

```
  title: Text('Add GPS'),
```

```
  value: _addGPS,
```

```
  onChanged: (value) => setState(() => _addGPS = value ?? false),
```

```
  controlAffinity: ListTileControlAffinity.leading,
```

```
),
```

```
SizedBox(height: 20),
```

```
/// Submit Button
```

```
ElevatedButton(
```

```
  child: Text('Book Now'),
```

```
  onPressed: () {
```

```
if (_formKey.currentState!.validate()) {  
  _formKey.currentState!.save();  
  
  // Show Summary  
  showDialog(  
    context: context,  
    builder: (_) => AlertDialog(  
      title: Text('Booking Confirmed'),  
      content: Column(  
        mainAxisAlignment: MainAxisAlignment.min,  
        crossAxisAlignment: CrossAxisAlignment.start,  
        children: [  
          Text('Name: $_name'),  
          Text('From: $_pickupLocation'),  
          Text('To: $_dropLocation'),  
          Text('Car Type: $_carType'),  
          Text('Payment: $_paymentOption'),  
          Text('Add Insurance: ${_addInsurance ? 'Yes' : 'No'}'),  
          Text('Add GPS: ${_addGPS ? 'Yes' : 'No'}'),  
        ],  
      ),  
      actions: [  
        TextButton(  
          onPressed: () => Navigator.pop(context),
```

```
        child: Text('OK'),  
      ),  
    ],  
  ),  
);  
}  
},  
)  
],  
,  
,  
,  
,  
,  
,  
);  
}  
}
```

Step 4 – Run your App

Flutter run

Output:

localhost:51604

Car Booking Form



Full Name

Pickup Location

Drop Location

Select Car Type

Payment Option:

☐ Cash

☐ Online

☐ Add Insurance

☐ Add GPS

Book Now

localhost:51604

Car Booking Form



Full Name

Enter name

Pickup Location

Enter pickup location

Drop Location

Enter drop location

Select Car Type

Please select a car type

Payment Option:

☐ Cash

☐ Online

☐ Add Insurance

☐ Add GPS

Book Now

localhost:51604

Car Booking Form



Full Name

SUCHITA PATEL

Enter name

Pickup Location

ANAND

Enter pickup location

Drop Location

VADODARA

Enter drop location

Select Car Type

SUV

Please select a car type

Payment Option:

☒ Cash

☐ Online

☒ Add Insurance

☐ Add GPS

localhost:51604

Car Booking Form



Full Name

SUCHITA PATEL

Pickup Location

ANAND

Drop Location

VADODARA

Select Car Type

SUV

Payment Option

☒ Cash

☐ Online

☒ Add Insurance

☐ Add GPS

Book Now

Booking Confirmed

Name: SUCHITA PATEL

From: ANAND

To: VADODARA

Car Type: SUV

Payment: Cash

Add Insurance: Yes

Add GPS: No

OK